

## 6. Low Level Control

The agent follows the primitives using a **PI control**.

Define the error:

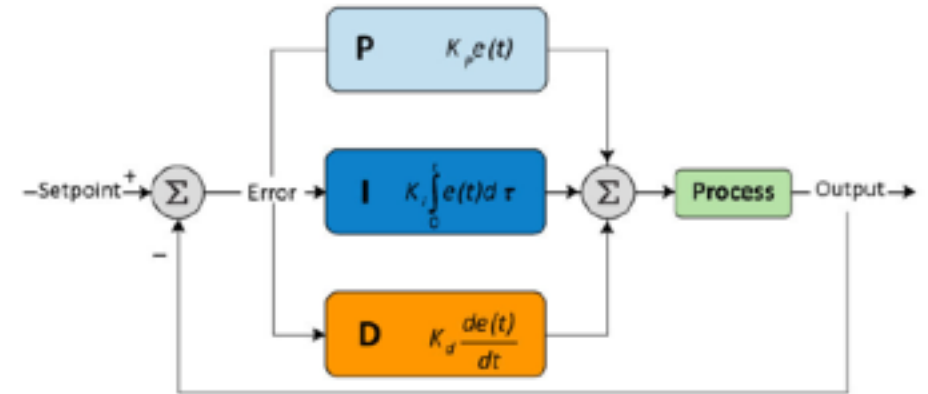
$$e = a_{req} - a;$$

Compute the integral of the error:

$$e_{int} = e_{int} + e \cdot dt$$

Requested pedal:

$$Req_{ped} = e \cdot P_{gain} + e_{int} \cdot I_{gain}$$



### Wind - Up problem:

If the integral term accumulates a **large** error (i.e. when the vehicle stops at the traffic light), it can cause **instability** or slow convergence to the target (i.e. **not** moving at green light) → so, the integral error is **imposed** to be **zero** when a velocity lower than few  $m/s$  is reached.



UNIVERSITÀ  
DI TRENTO  
Dipartimento di  
Ingegneria Industriale

**Master Course:** Mechatronics Engineering  
INTELLIGENT VEHICLES and AUTONOMOUS DRIVING

# Autonomous Driving System for Traffic Light

Professor

Rosati Papini Gastone Pietro

Student

Murgia Edoardo, Mat: 247210

ACADEMIC YEAR  
2024/2025